# MILITARY SPECIFICATION

## GAGE, SURFACE PLATE FLATNESS CHECKING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE.

1.1 Scope. This specification covers a gage for checking surface plate wear, including a master reference gage calibrator, complete with protector cases and covers.

## 2. APPLICABLE DOCUMENTS.

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

| | |
|---|---|
| PPP-T-97 | Tape, Pressure-Sensitive Filament Reinforced |
| PPP-B-601 | Boxes, Wood, Cleated Plywood |
| PPP-C-843 | Cushioning Material, Cellulosic |

Military

| | |
|---|---|
| MIL-P-116 | Preservation – Packaging, Methods of |
| MIL-G-10944 | Gages, Dimensional Control |
| MIL-I-18422 | Indicators, Dial and Accessories |

STANDARDS

Military

| | |
|---|---|
| MIL-STD-129 | Marking for Shipment and Storage |
| MIL-STD-130 | Identification Marking of U.S. Military Property |

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.



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Uniform Classification Committee (UCC)

Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, 202 Uniform Station, 516 West Jackson Boulevard, Chicago, Illinois 60606)

3. REQUIREMENTS.

3.1 First article. Unless otherwise specified (see 6.2), this specification provides for first article testing and approval. A first article comprises a preproduction model or initial production sample; approval involves testing and evaluating the first article for conformance with the requirements herein before, or in the initial stage of, production under a contract. First article approval shall be in accordance with Section 4.

3.2 Design. Unless otherwise required by this specification, gages shall comply with requirements of MIL-G-10944. The gage assembly shall include a calibrated straight edge and a dial indicator.

3.2.1 Interchangeability. All parts shall be manufactured to such standards as will permit replacement or adjustment without modification of component parts.

3.2.2 Material. All parts used in fabrication of the gage shall be new and of the quality necessary to produce a gage assembly to meet the requirements described herein.

3.3 Construction. The gage assembly shall be complete with all necessary components and calibrating equipment so that when assembled and calibrated it will accurately perform the gaging operation for which it is designed. Construction shall be free from any characteristics or defects that will prevent the gage assembly from passing any of the examinations and tests described in Section 4.

3.3.1 Castings and forgings. All castings and forgings shall be free from defects. In no event shall such processes as welding, peening, plugging, filling with cold solder or metallic pastes be used for reclaiming any part of the gage assembly and calibrating equipment.

3.3.2 Fastening devices. All screws, pins and bolts shall be installed with means for preventing loss of tightness. All such parts, when subject to removal or adjustment, shall not be swaged, peened, staked or otherwise permanently deformed.

3.3.3 Surface finish. All surfaces shall be smooth and free from burrs and sharp edges. Surfaces that are subject to corrosion when exposed to normal use shall be protected against such corrosion in a manner that will in no way prevent compliance with this specification. The protective coating shall not loosen, chip, crack or scale with normal use and age.



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3.4 Components.

3.4.1 Dial indicator. The dial indicator shall be either an American Gage design standard or an equivalent gage design standard which will meet or exceed the requirements described herein and as otherwise indicated in Military Specification MIL-I-18422. Dial graduations on the indicator shall be for not greater than 20 millionths of an inch. Measurement accuracy of the indicator shall be within one graduation of the dial indicator. Total full scale dial range of the indicator shall be not less than plus or minus one thousandth of an inch. Full scale deflection of the indicator shall not require more than 4 ounces of force applied to the contact point. The indicator or the indicator mounting shall have minute adjustment provisions for zeroing dial indications (a loading is allowable on the spindle when this adjustment is made). The dial indicator shall have the capability of indicating surface variations of 0.000020 inches.

3.4.1.1 Contact point. Unless otherwise specified (see 6.2), the contact point shall be either a ruby or a material of equal or better wear resistance and hardness when compared to ruby.

3.4.2 Indicator mounting bar or block. The bar or block shall be designed to provide rigid reliable support for the indicator. An extension handle or other suitable means shall be provided to minimize temperature change and possible erroneous measurement as a result of handling the indicator mounting block. The bar or block shall be provided with a minute adjustment feature for positioning the indicator.

3.4.3 Calibrated straight-edge. The straight-edge shall be used in conjunction with the block mounted indicator. The straight-edge shall be equipped with not less than one fixed support and one adjustable support. The height of the fixed support shall correspond with the height of the block mounted indicator. The adjustable support shall be provided with a minute precise adjustment feature for positioning both ends of the straight-edge in a reference plane that is parallel to the surface being checked.

3.4.3.1 Straight-edge sizes and accuracies. The overall guaranteed flatness accuracies shall be within the tolerances specified in Table I for the respective straight-edge sizes shown. The size of straight-edge required shall be as specified (see 6.2). This size is the utile length of the straight-edge that is applicable.

Table I

| Straight-edge Length (Inches) | Flatness Tolerances (Inches) |
|---|---|
| 24 | 0.000050 |
| 36 | 0.000070 |
| 48 | 0.000130 |
| 60 | 0.000180 |



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3.5 Cases and covers. Protective cases and covers are required and shall be constructed in accordance with manufacturers standard practice for the items involved.

3.6 Nameplate. A corrosion resistant metal nameplate, permanently and legibly marked with the following information in accordance with MIL-STD-130, shall be securely attached to each gage assembly.

Nomenclature
Manufacturer's name
Manufacturer's model number
Manufacturer's serial number
Date of Manufacture
Contract or Order number
Federal Stock Number
U.S.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection shall be classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. First article inspection shall be applied to the preproduction sample submitted in accordance with 3.1. Failure of the preproduction sample to pass the examination or one or more preproduction tests shall be cause for rejection. First article inspection shall consist of the examination in 4.3 and all test in 4.4.

4.2.2 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Failure of the gage to pass the examination shall be cause for rejection. Quality conformance inspection shall consist of the following:

a. Examination (4.3).
b. Tests (4.4.1 and 4.4.2).
c. Inspection of preservation, packaging, packing and marking for shipment and storage (4.5).



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4.3 Examination. The gage assembly and equipment shall be examined for design, construction, materials, components, and workmanship to determine compliance with the requirements of this specification. Any deviation from these requirements shall be cause for rejection.

4.4 Tests.

4.4.1 Indicator inspection. Each indicator shall be inspected to assure proper function of all adjustments and moving parts. The spindle shall be fitted with the specified contact point (see 3.4.1.1) and shall be actuated through its entire range of travel without evidence of sticking or malfunction.

4.4.2 Straight-edge accuracy check. Each straight-edge shall be calibrated under Metrology Laboratory conditions to insure the surface accuracy requirements specified in 3.4.3.1 and Table I. Failure of the gaging reference surface to conform to the tolerance accuracy specified shall be cause for rejection.

4.4.3 Functional test. The completely assembled gage shall be calibrated and subjected to the following test. Surface measurements shall be taken perpendicular to a straight line on a flat surface not less than the utile surface length of the straight-edge required. The first measurement shall be taken one inch inside of extreme utile surface ends of the applicable straight-edge and at two inch intervals for the remaining straight-edge utile surface length. The exact dial reading for each interval shall be recorded. It is imperative that each specific measurement interval be clearly defined. The measuring sequence shall be repeated not less than three times and the dial readings compared for repetitive accuracy. Deviations of more than one dial graduation at each measurement interval shall constitute failure of the test. There is no requirement for the testing surface used in this calibration to be delivered with the item required by this specification.

4.5 Inspection of preservation, packaging, packing and marking for shipment and storage. Preservation, packaging, packing and marking for shipment and storage shall be inspected to determine conformance with the requirements specified in Section 5 of this specification.

5. PERPARATION FOR DELIVERY

5.1 Preservation and Packaging. Preservation and packaging shall be Levels A, B or C as specified (see 6.2).

5.1.1 Levels A and B. Gages shall be packaged, one each, Method III in accordance with MIL-P-116, as follows:

5.1.1.1 Indicator. The dial indicator shall be placed in a protective case. It shall then be enclosed in a moisture proof bag. Bag closure may be made by any suitable means.

5.1.1.2 Other components. Remaining components, such as the straight-edge, and straight-edge supports shall be individually placed in their respective carrying case.



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5.1.1.3 Packing. Pack all parts as indicated in 5.1.1.1 and 5.1.1.2 into the manufacturers carrying case, when provided. Voids within the carrying case that could allow damage to gage components shall be filled with cushioning material conforming to PPP-C-843.

5.1.1.4 Case. The manufacturers carrying case shall be closed and secured or latched in the closed position. Further securing of the container shall be accomplished by using three bands of tape, pressure sensitive, adhesive, filament reinforced in accordance with PPP-T-97, (placed girthwise, equally spaced from ends).

5.1.2 Level C. Gages and accessories shall be packaged in accordance with the manufacturers standard commercial practice.

5.2 Packing. Packing shall be Levels A, B, or C as specified (see 6.2).

5.2.1 Level A. Gages preserved and packaged as specified in para 5.1 shall be overpacked into an exterior snug fitting, overseas, wood cleated, plywood box, PPP-B601, style optional. Exterior containers insofar as practicable shall be of a uniform size and minimum cube consistant with the protection required. Gross weight shall not exceed specification requirements. Container closure shall be in accordance with container specification and appendix thereto.

5.2.2 Level B. Gages preserved and packaged as specified in Paragraph 5.1 shall be packed the same as for Level A, except that containers shall be domestic type.

5.2.3 Level C. Gages preserved and packaged in accordance with paragraph 5.1shall be packed to assure carriers acceptance and safe delivery to destination at the lowest rates. Containers shall meet Uniform Freight Classification Rules or regulations or other common carriers as applicable to the mode of transportation.

5.3 Marking. In addition to any special markings as required by contract or order (see 6.2), marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The intended use of the surface plate flatness checking gage covered by this specification is for periodic checking of surface plate wear and condition at prescribed time intervals and at times when excessive wear is suspected.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. First article inspection, if different (see 3.1 and 4.2.1).
c. Contact point material, if different (see 3.4.1.1)
d. Straight-edge utile surface length required (see 3.4.3.1 and Table I).
e. Selection of applicable levels of preservation, and packaging (see 5.1).



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f. Selection of applicable level of packing (see 5.2).
g. Markings required (see 5.3).

Custodians:
Air Force – 84
Army - MU

Reviewers:
Army – AT, ME, MU

Preparing Activity:
Air Force – 84

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